

Applications of Generative AI in Architectural Design Education: A Systematic Review and Future Insights

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Abstract

This study reviews the current applications of generative artificial intelligence (GenAI) in architectural design education using the PRISMA framework. It compares these applications across the different design stages, namely the pre-design, concept generation, design development, and design production, to identify the current state of evidence and conceptual discussions reported in the literature. The study also discusses the associated opportunities and challenges in this regard. The findings indicate that there is a growing interest in integrating GenAI into architectural design education, especially in the early design stages. However, one of the most significant gaps in this regard lies in the lack of empirical evidence on the long-term impacts of GenAI on students' critical thinking and problem-solving skills. Future research is needed to explore the integration of GenAI throughout the entire design process, including design development and refinement. There is also a need to incorporate the relevant ethical guidelines for AI-generated content into academic quality assurance systems and to strengthen institutional preparedness through targeted training and policy development.

Keywords: architecture; design; education; engineering; generative artificial intelligence

1. Introduction

Architecture is a discipline that shapes spaces to meet human needs, combining creativity, technical needs, and aesthetic values [1]. Architectural design, as a subset of this field, involves solving complex spatial problems [2]. It is a decision-making process that integrates scientific and technological tools to deliver efficient and innovative design solutions. Architectural education is a unique discipline that demands high levels of creativity and distinct pedagogical approaches. It encourages spatial thinking, critical reflection, and the ability to translate abstract concepts into physical forms. Over time, architectural pedagogy has evolved from traditional methods to more interactive and interdisciplinary approaches to respond to the contemporary shifts in technology and societal needs. This necessitates offering academic programs that combine knowledge and high-level thinking competencies with hands-on skills to prepare students for the current and future job market. These qualities are foundational to architectural training. They also reflect the broader aim of the discipline to meet human needs through meaningful design approaches [1,3].



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Design studio is one major component of any educational program in architecture. It offers an interactive learning environment that fosters hands-on engagement, reflective practice, and iterative development through experimentation and feedback given to students collectively and individually. This fosters ongoing dialogue among students and instructors, allowing ideas to evolve through brainstorming and critique [4]. This environment is also crucial for addressing real-life architectural challenges and developing the critical and technical skills necessary for architectural practice. This also cultivates interpersonal learning and peer exchange, which are crucial in addressing real-world challenges and developing students' competencies. Compared to other subjects in architecture, the design studio has the highest number of weekly contact hours. On the other hand, the rest of the subjects, such as building technology, architectural history and theory, and architectural communication, are designed to support and enrich the learning that occurs within the studio [3]. Students communicate their ideas in the design studio using diverse media, including sketches, physical and digital models, and visual presentations.

However, despite its central role, traditional design and communication methods are increasingly criticized for their rigid structures, which may limit the ability of students to explore innovative design approaches that reflect the complexities of contemporary practice [5]. This has prompted a growing call among educators to reform the architectural design studio pedagogy by integrating new technologies that better prepare students for the evolving demands of the architectural profession [4]. One main driver in this regard is the emergence of Artificial Intelligence (AI) and the rise of digital and AI-driven technologies. These disruptive technologies have fundamentally challenged the traditional pedagogical methods in architectural education. This necessitates the need to revise how design is taught, practiced, and conceptualized. Among the most transformative technologies in this regard is the GenAI, which introduces new paradigms of creativity, process automation, and interaction between design teams and machines. Thus, this study focuses on GenAI, rather than AI, because GenAI directly influences ideation and iterative design thinking through content creation. As a result, its pedagogical implications differ significantly from those of analytical or decision-support AI.

Despite growing attention to the impact of GenAI use on architecture, there is a need to systematically map GenAI applications used in architectural design education with the different design-process stages. Accordingly, this review contributes a stage-level synthesis that maps GenAI applications across design stages, highlighting where learning-related evidence is reported and where claims remain primarily conceptual. This review aims to synthesize recent studies on GenAI in architectural design education, organize and compare reported applications across key design stages, namely the pre-design analysis, concept generation, design development, and design production, and identify opportunities, challenges, and research gaps in this regard, as discussed below.

2. Materials and Methods

This research involved a systematic review to map and synthesize existing studies on the application of artificial intelligence in architectural design education and the design stages. The review process followed the PRISMA 2020 systematic review approach to ensure transparency, rigor, and reproducibility. This study is a PRISMA-guided systematic review, aiming to synthesize and map GenAI applications and evidence across architectural design stages rather than quantify effect sizes. The process included a critical and relevant keyword search, careful database selection, and thorough screening.

2.1. Keyword Search and Databases

A comprehensive keyword search was conducted in the academic database Scopus; in addition, a search was conducted in the search platform Google Scholar to identify further relevant scholarly work. The Boolean search strings combined terms included the following combinations:

("Artificial Intelligence" OR "Generative AI" OR AI) AND ("Design Studio" OR "Architectural Design" OR "Design Process" OR "Architectural Process"). To ensure the inclusion of all relevant studies, additional terms were added to the search string, such as ("Architectural Design Education" OR "Architectural Education" OR Architecture). The search terminology was carefully constructed to capture a broad and relevant set of studies focusing on the intersection of AI and architectural design education. The initial search yielded 652 records.

2.2. Identification and Screening

To refine the results, two stages of Automatic Database Exclusion (ADE) were applied:

- ADE#1: Records were filtered to include only English-language publications from the 2020–2025 timeframe, resulting in 302 documents.
- ADE#2: The results were further narrowed to peer-reviewed journal articles and review papers only, excluding conference papers, proceedings, and other non-article documents. This step reduced the dataset to 124 documents.

Furthermore, an initial screening was conducted based on titles, abstracts, and keywords, which resulted in retaining 52 records relevant to the study topic. Then, a full screening was manually conducted using the following inclusion criteria:

- Focus on AI or GenAI within architecture or architectural education.
- Relevant discussion of AI applications across the design stages: pre-design, conceptual design, design development, or production.
- Insights into pedagogical shifts, studio applications, or learning outcomes.

In addition to the above, further research was conducted using Google Scholar, which initially identified 65 documents after full-text screening. After removing duplicates already found in Scopus, 8 additional unique records were retained, resulting in a final set of 45 unique publications considered for the study. However, studies focusing purely on unrelated computational methods without a clear link to education or architectural practice were excluded. Figure 1 presents the PRISMA flow chart documenting the whole selection process.

2.3. Eligibility and Final Inclusion

Eligible full texts were further assessed to ensure alignment with the scope of the review. In this context, architectural design education studies refer to the studies that discuss GenAI applications within architectural teaching or learning settings. Professional and urban design studies were included only when their insights were explicitly transferable to design pedagogy and aligned with curriculum-relevant design stages. Relevant citations were manually traced to capture additional sources. After final screening, unrelated, non-open-access, and retracted papers were excluded, resulting in 45 studies included in the systematic review. A thematic mapping and synthesis of the final dataset was conducted to organize insights by:

- The stages of the architectural design process.
- GenAI tools used and their educational applications.
- Gaps and opportunities for future research.

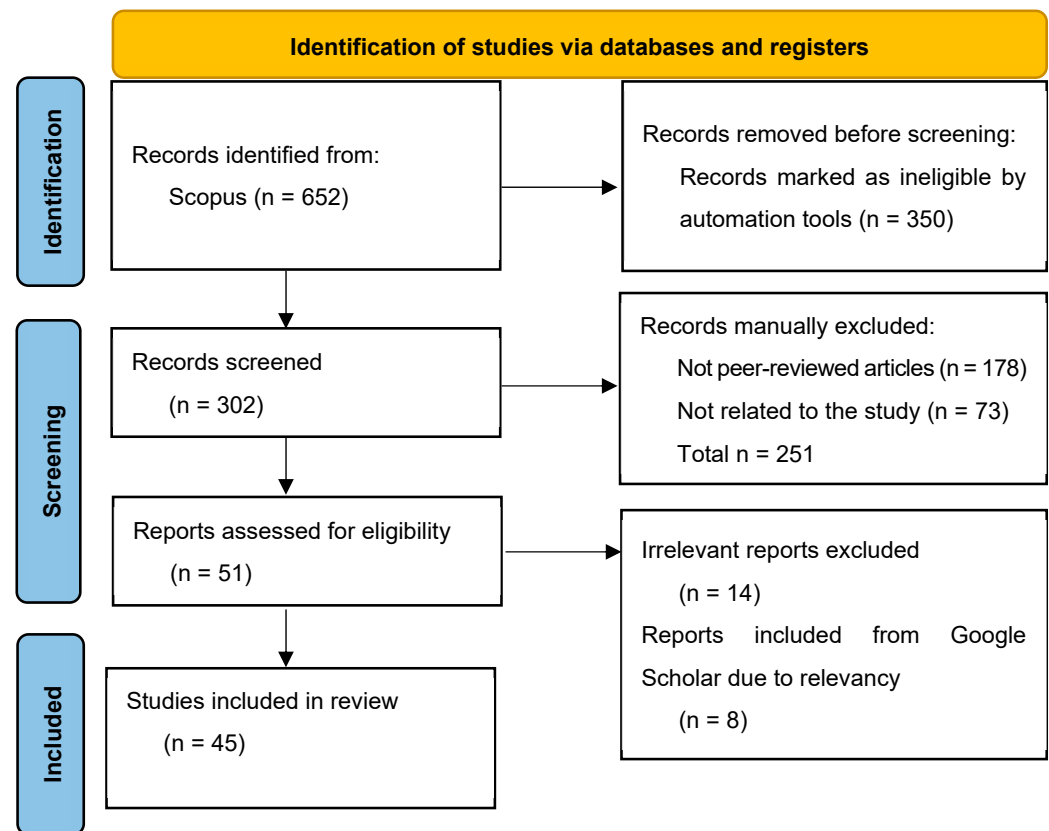


Figure 1. PRISMA flow diagram of the review process.

Included studies were classified into three evidence types: empirical, conceptual, or technical, based on the primary contribution of each study. This classification was used to support synthesis across design stages and to distinguish between demonstrated educational effects and conceptual or tool-oriented contributions. The final set of studies was analyzed through a thematic synthesis approach. Key information was extracted from each publication and categorized under thematic areas that correspond to the structure of this review, including the traditional architectural design process and its limitations, the integration of GenAI in architectural design education, and the AI tools and applications used in this regard, as shown in Table 1. Also, Table 2 demonstrates a thematic mapping that was conducted to map the reviewed sources to the study scope, organize findings to enhance clarity and coherence, and identify possible gaps in the literature.

Table 1. List of articles considered in this review.

No.	Citation	Year	Country of Publication	Scope of the Study	Evidence Type
1	Al-Soufi & El Shafie [6]	2025	KSA	Examines the current implementation of AI-assisted design tools in architecture by analyzing their practical application through the design of a residential building in Riyadh.	Empirical
2	Belaroussi & Martín-Gutierrez [7]	2025	France/Spain	Investigates how ChatGPT's interpretation of architectural ambiance compares to human perception, analyzing similarities, differences, and potential implications for AI-supported architectural design evaluations.	Empirical

Table 1. *Cont.*

No.	Citation	Year	Country of Publication	Scope of the Study	Evidence Type
3	Braiden et al. [8]	2025	Canada/USA	Presents findings from a professional survey of landscape architects in North America to examine current uses, perceptions, and future prospects of AI in landscape architecture practice.	Empirical
4	Cheung et al. [9]	2025	China	Explores three approaches for integrating conversational, agentic, and multimodal AI tools into early-stage performative architectural design processes.	Empirical
5	Deregibus [10]	2025	Italy	Develops a systemic framework that highlights the strategic use of keywords to organize, guide, and enhance decision-making within the architectural design process.	Conceptual
6	El Moussaoui [11]	2025	Italy	Examines how the integration of AI is transforming traditional architectural practice processes, focusing on the evolving relationship between designers and AI tools.	Conceptual
7	Karadağ & Ozar [12]	2025	Turkey	Explores the collaborative potential between AI tools and human designers in the conceptual design phase within architectural design studios.	Empirical
8	Lekesiz & Müezzinoğlu [13]	2025	Turkey	Presents an approach to integrating AI-supported learning in architectural education through a case study focusing on speculative space design.	Empirical
9	Rodriguez et al. [14]	2025	Peru	Examines how artificial intelligence and virtual reality technologies influence users' perception and experience of architectural design.	Empirical
10	Schroth & Maier [15]	2025	Germany	Explores methods for integrating generative artificial intelligence into the landscape architecture design process to enhance creativity and efficiency.	Empirical
11	Shokry [16]	2025	Egypt	Investigates how AI influences urban design practices, with a focus on its potential to shape planning, analysis, and decision-making processes.	Conceptual
12	Asfour [17]	2024	KSA	Explores the potential impacts of AI on architectural design education, including design studio practice, student creativity, and the role of educators.	Conceptual
13	Bassey et al. [18]	2024	UK/USA	Examines how AI techniques improve the accuracy and efficiency of life cycle assessment (LCA) for renewable energy systems in the built environment.	Technical
14	Cao et al. [19]	2024	China	Compares the use of artificial intelligence and virtual reality in sustainable architecture education, focusing on how these technologies can reinterpret traditional design concepts and support learning.	Empirical

Table 1. *Cont.*

No.	Citation	Year	Country of Publication	Scope of the Study	Evidence Type
15	Çınar Kalenderoğlu & Demiröz [20]	2024	Turkey	Analyzes the integration of text-to-image AI tools in architectural design education through insights gained from a design studio experience.	Empirical
16	Cudzik & Nyka [21]	2024	Poland	Examines how AI tools support architectural education through a green campus development project, highlighting their role in conceptual design, sustainability analysis, and design production.	Empirical
17	Fareed et al. [22]	2024	UAE/Egypt	Investigate how AI image generators can be used as educational tools to support teaching architectural history by visualizing historical styles and concepts.	Empirical
18	Golkarian [23]	2024	Turkey	Explores how AI-driven generative ideation can support the development of architectural spaces inspired by Iranian traditional urban forms, using a case study to demonstrate concept generation and spatial design enhancement	Empirical
19	Günaydın et al. [24]	2024	Turkey	Examines the role of artificial intelligence as a pedagogical tool to enhance teaching and learning processes in architectural design education.	Conceptual
20	Jin et al. [25]	2024	Malaysia	Analyzes how traditional architectural elements influence the outcomes of AI-generated designs using computational methods.	Technical
21	Jo et al. [26]	2024	Korea/USA	Explores the use of GenAI models trained on local identity to produce early photorealistic renderings of building façades in the design process.	Technical
22	Karadag & Yıldız [27]	2024	Turkey	Provides an overview of recent AI innovations in architecture, discusses implementation challenges, and analyzes ethical implications.	Conceptual
23	Karimi et al. [28]	2024	Turkey/Iran/Italy/USA	Demonstrates the use of deep learning and reinforcement learning to optimize building energy performance during the architectural design process.	Technical
24	Khogali [29]	2024	KSA	Examine how integrating AI tools affects design development and learning outcomes in an architecture college setting, particularly in design studios.	Empirical
25	Li et al. [30]	2024	UK	Demonstrates how GenAI models can transform simple sketches into detailed architectural floor plans and 3D models, highlighting the workflow and potential of AI-assisted sketch-to-architecture processes.	Technical
26	Maksoud et al. [31]	2024	UAE	Examines the integration of an image-GenAI tool into the creative brainstorming process for developing a conceptual form of Safavid mosque architecture.	Empirical
27	Montenegro [32]	2024	Portugal	Provides an integrative analysis of text-to-image AI systems in architectural design education, focusing on their pedagogical innovations and impact on creative design processes.	Conceptual

Table 1. *Cont.*

No.	Citation	Year	Country of Publication	Scope of the Study	Evidence Type
28	Paananen et al. [33]	2024	Finland	Investigates the use of text-to-image generation tools to support ideation in architectural design processes.	Empirical
29	Płoszaj-Mazurek & Ryńska [34]	2024	Poland	Explores how AI combined with Building Information Modeling (BIM) can support low-carbon architectural design by improving life cycle assessment tools and processes.	Technical
30	Sindhu Devi & Maruthuperumal [35]	2024	India	Provides an overview of ChatGPT, its capabilities as a language model, and its potential uses and limitations across various fields.	Conceptual
31	Xu et al. [36]	2024	USA	Reviews how GenAI supports the autonomous creation of urban data, scenarios, designs, and 3D models to advance smart city development and urban design processes.	Conceptual
32	Zwangsleitner et al. [37]	2024	Germany	Examines the role of AI as a tool to support and enhance the landscape architecture design process.	Empirical
33	Bölek et al. [38]	2023	Turkey	Provides a comprehensive overview of how AI technologies are being applied across architectural design phases, with a focus on tools, methods, and potential benefits.	Conceptual
34	Caliskan [39]	2023	Turkey	Investigates the potential, challenges, and limitations of using ChatGPT as a knowledge source for shaping tasks in an architectural design studio.	Empirical
35	Derevyanko & Zalevska [40]	2023	Ukraine	Compares the features, capabilities, and educational applications of Midjourney, Stable Diffusion, and DALL-E for supporting design students' creative work and visual outputs.	Conceptual
36	Fernberg et al. [41]	2023	/USA	Explores the use of AI-powered image generators for creating 2D asset libraries to support architectural and design workflows.	Technical
37	Desouki et al. [42]	2023	Egypt	Explores the dual role of revolutionary AI design solutions in architecture, analyzing whether they offer opportunities or pose risks to traditional design practice and creativity.	Conceptual
38	Meron & Tekmen Araci [43]	2023	Australia	Assesses the feasibility and effectiveness of using ChatGPT-4 as a collaborative virtual colleague to assist educators in developing postgraduate design studio courses.	Empirical
39	Milošević et al. [44]	2023	Serbia	Explores how AI tools automate and expand conceptual design explorations in architecture by generating diverse design compositions and solutions.	Technical
40	Rane [45]	2023	India	Examines how ChatGPT and comparable GenAI tools can be used in architectural engineering, highlighting their roles, benefits, and the challenges they pose for integration.	Conceptual
41	Tabrizi et al. [46]	2023	Australia	Examines how AI tools can support teaching architecture students about circular design principles and conducting life cycle assessments to promote sustainability.	Empirical

Table 1. Cont.

No.	Citation	Year	Country of Publication	Scope of the Study	Evidence Type
42	Yudhanta & Hadinata [47]	2023	Indonesia	Investigates how computational methods and AI tools support tasks in the architectural pre-design phase, demonstrated through a residential design case study.	Technical
43	Baduge et al. [48]	2022	Australia	Reviews the integration of AI, machine learning, and smart vision technologies to improve efficiency, safety, and sustainability in the building and construction sector under Industry 4.0 frameworks.	Conceptual
44	Ploennigs & Berger [49]	2022	Germany	Examines how text-to-image GenAI tools can be integrated into architectural design workflows to support concept generation, visualization, and creative exploration.	Technical
45	Castro Pena et al. [50]	2021	Spain	Provides a comprehensive review of how artificial intelligence is applied to support and enhance the conceptual design stage in architectural practice.	Conceptual

Table 2. Mapping of the reviewed studies with the different architectural design process stages.

No.	Source	Architectural Design Process Stages			
		Pre-Design Analysis	Conceptual Design	Design Development	Design Production
1	Al-Soufi & El Shafie [6]	✓			
2	Belaroussi & Martín-Gutierrez [7]	✓			
3	Braiden et al. [8]	✓			
4	Cheung et al. [9]	✓		✓	
5	Deregibus [10]		✓		
6	El Moussaoui [11]			✓	✓
7	Karadağ & Ozar [12]		✓		
8	Lekesiz & Müezzinoğlu [13]		✓		
9	Rodriguez et al. [14]	✓			
10	Schroth & Maier [15]		✓		
11	Shokry [16]				✓
12	Asfour [17]		✓		
13	Bassey et al. [18]				✓
14	Cao et al. [19]	✓			
15	Çınar Kalenderoğlu & Demiröz [20]	✓	✓		
16	Cudzik & Nyka [21]		✓		✓
17	Fareed et al. [22]		✓		
18	Golkarian [23]				✓
19	Günaydın et al. [24]				✓
20	Jin et al. [25]		✓		
21	Jo et al. [26]				✓
22	Karadag & Yıldız [27]			✓	
23	Karimi et al. [28]			✓	
24	Khogali [29]			✓	✓
25	Li et al. [30]			✓	
26	Maksoud et al. [31]		✓		
27	Montenegro [32]			✓	
28	Paananen et al. [33]		✓		

Table 2. *Cont.*

No.	Source	Architectural Design Process Stages			
		Pre-Design Analysis	Conceptual Design	Design Development	Design Production
29	Płoszaj-Mazurek & Ryńska [34]				✓
30	Sindhu Devi & Maruthuperumal [35]	✓			
31	Xu et al. [36]			✓	
32	Zwangsleitner et al. [37]		✓		
33	Bölek et al. [38]			✓	
34	Caliskan [39]	✓			
35	Derevyanko & Zalevska [40]		✓		
36	Fernberg et al. [41]				✓
37	Desouki et al. [42]		✓		
38	Meron & Tekmen Araci [43]	✓			
39	Milošević et al. [44]			✓	
40	Rane [45]	✓			✓
41	Tabrizi et al. [46]				✓
42	Yudhanta & Hadinata [47]		✓		
43	Baduge et al. [48]				✓
44	Ploennigs & Berger [49]		✓		
45	Castro Pena et al. [50]		✓		

3. Findings

GenAI is increasingly transforming architectural design by expanding creative possibilities and streamlining architectural workflow [51]. It enables architects to generate multiple design options quickly based on specific parameters, improving efficiency and innovation compared to traditional methods [52]. These tools are also entering architectural education, offering new ways to support learning and design thinking [29]. While GenAI presents many opportunities, it also raises concerns about over-reliance and the potential loss of human creativity [53]. This section highlights how AI is currently being applied within the different stages of the architectural design process, as presented in Table 2. It highlights its capabilities, current applications, and the opportunities and challenges that exist in this regard.

3.1. The Architectural Design Process

The architectural design process is inherently iterative, requiring students to refine their ideas through feedback and critique. This enables exploration of multiple design alternatives and fosters innovation and problem-solving skills. Creativity in this context involves generating new ideas and identifying unseen connections within given constraints. This is practiced in architecture design studios, where students engage in dialogues about their projects to translate conceptual ideas into visual forms. This collaborative nature of design studios offers peer learning and instructor guidance, which enhances students' ability to reason and communicate decisions effectively [54]. Ultimately, this cycle of creation and critique strengthens critical thinking skills needed to address architectural complexities. The architectural design process is approached through various models, which highlight the importance of a structured yet flexible design process. This process is commonly divided into several stages, each of which includes several tasks [55–57]. The design process is commonly divided into the following four stages (Figure 2):

- The pre-design analysis phase, which involves some tasks such as programming, design data collection and site analysis.

- The conceptual design phase, which involves crafting and developing the initial conceptual model, including the design philosophy.
- The design development phase, where the project's characteristics, such as spatial organization and circulation, are clarified using two-dimensional (2D) drawings and three-dimensional (3D) massing models.
- The design production phase, which includes refinement of the 2D drawings, 3D rendering of the project, and preparing presentation materials to effectively communicate the design.

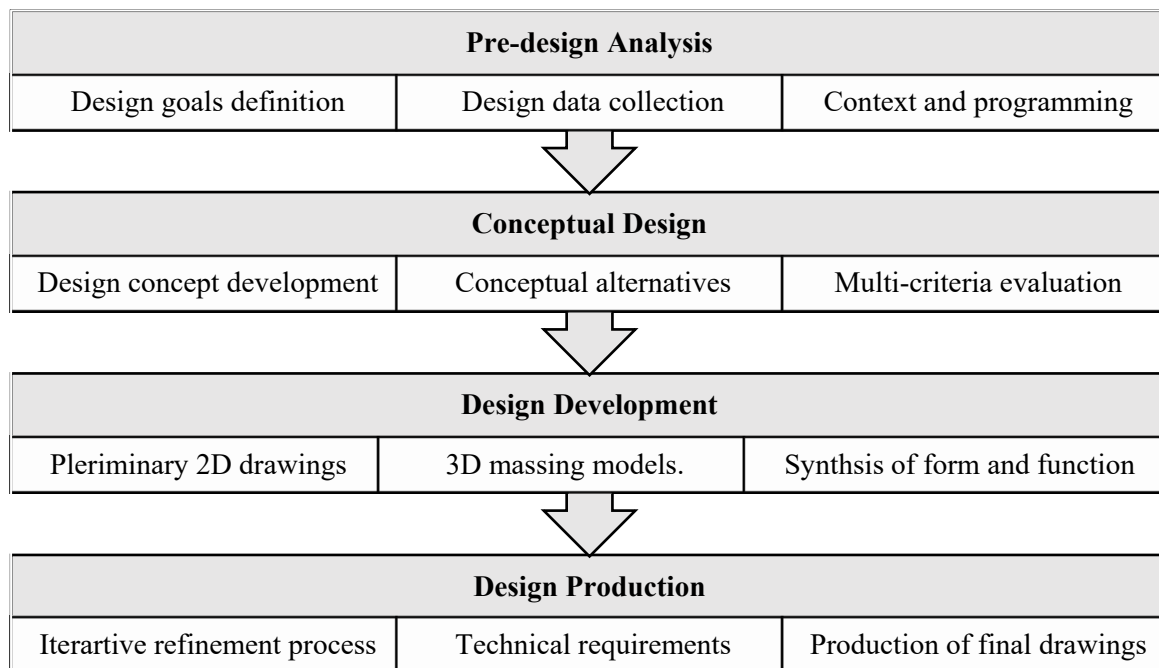


Figure 2. Main stages of a typical architectural design process.

However, implementation of these design tasks in architectural design education faces several challenges that limit its effectiveness in preparing students for the complexities of contemporary professional practices. One primary issue in this regard lies in the reliance of architecture schools on conventional and outdated teaching methods in the architectural design studio [58]. Additionally, traditional studios often place more emphasis on the final outcome than on the design process itself, discouraging experimentation and limiting creativity [59]. As Desouki et al. [5] highlighted, this can cause students to prioritize aesthetics over deeper problem-solving and conceptual development. To overcome these challenges, several approaches are proposed, including:

- Active learning methods to encourage critical thinking and problem-solving skills through interactive and participatory learning environments [5,59,60].
- Interdisciplinary collaboration to offer multidisciplinary knowledge and foster teamwork skills [61].
- Utilization of advanced digital tools and disruptive technologies to facilitate greater collaboration and innovation [62,63].

3.2. The Shift Towards Digitization in Architectural Design Education

The accelerated integration of advanced digital technologies into architectural design has significantly reshaped the discipline, driving remarkable advancements in efficiency and innovation. Students nowadays are increasingly accustomed to digital platforms in their daily lives, yet traditional design education often fails to leverage these tools

effectively. Consequently, architectural education that does not evolve alongside advancements in digital design and construction technology risks leaving students inadequately equipped for the demands of contemporary architectural practice [5]. Since the development of Computer-Aided Design (CAD), architectural design has gradually evolved into a digital practice.

Digitalization has enabled the integration of computational design, Virtual Reality (VR) and Augmented Reality (AR), simulation engines, and parametric modeling into architectural education [64]. This digital transition challenges existing pedagogical models while offering new methods for collaboration, experimentation, and visualization. The COVID-19 pandemic further accelerated this digital shift, compelling academic institutions to adopt remote and hybrid learning modes. This also compelled educators to rethink learning outcomes, assessment methods, and instructional strategies [65]. In parallel, recent educational discussions have highlighted the need to reconsider traditional studio pedagogy under rapidly evolving digital technologies. Accordingly, students are increasingly expected to develop digital fluency alongside traditional design thinking, including the informed use of AI within studio workflows [51]. Over the last decade, AI has undergone transformative integration into various sectors, including architecture. Defined as the ability of machines to replicate human cognitive functions, such as learning, reasoning, and problem-solving, AI has evolved from a theoretical concept introduced by John McCarthy in 1956 to a robust, interdisciplinary field impacting nearly all aspects of modern life [66,67]. In architecture education, AI's potential is particularly pronounced due to the field's inherent complexity and reliance on critical thinking, spatial visualization, and iterative creativity. The integration of AI into architectural design education presents an opportunity not only to enhance learning outcomes but also to support rethinking the architectural design process itself. Cooper [68] argued that AI can amplify creativity, streamline data analysis, and improve performance-based design strategies. Moreover, it enables students to understand and manage complex datasets while exploring architectural concepts in greater depth.

GenAI nowadays represents a paradigm shift in architectural education. GenAI refers to AI systems capable of producing new content, text, images, 3D models, or code by recognizing and replicating patterns within large datasets [69,70]. It operates through an ecosystem of technologies, including Deep Learning (DL), Machine Learning (ML), Natural Language Processing (NLP), Artificial Neural Networks (ANN), and Large Language Models (LLMs), all contributing to its generative capacities [71,72]. GenAI can support both the creative and analytical dimensions of design, aligning closely with the discipline's core values [52]. It may empower students to rapidly test a broader range of design solutions, optimize outcomes, and simulate environmental performance metrics [29], as discussed in the following section. Importantly, within increasingly digital studio settings, this shift is pedagogically disruptive because content generation can influence core learning activities (e.g., ideation, iteration, and representation) rather than only digitizing documentation or visualization [65,69–72].

3.3. The Use of GenAI in Architectural Design Education

AI is increasingly influencing architectural practice, prompting discussions about the need for its integration into architectural education. Yildirim [73] argued that the adoption of AI tools in education highlights the need to question the efficiency of the traditional pedagogical approaches. Ceylan [74] highlighted that architecture is one of the most affected disciplines by the advancement of AI technology because of its nature based on design skills. Başarır [75] emphasized that dedicated experimental AI courses in architectural schools help students comprehend AI technologies and develop the skills needed to apply them throughout the design process. However, several studies have suggested that a balance

must be maintained between technology-based and human-centered design approaches, ensuring these are observed without compromising the development of manual skills and competencies [65,76]. The following sections discuss the potential use of GenAI in the different architectural design process stages.

3.3.1. Pre-Design Analysis

At this stage, AI chatbots powered with LLM, such as ChatGPT, DeepSeek, and Bard, can potentially assist in gathering project information, analyzing site constraints, and generating preliminary design ideas [45]. They may also help review environmental factors, zoning regulations, and user preferences to inform the design process. They can also provide human-like conversations, which can answer questions and assist with tasks like programming and design data collection [35]. Within a context-based architectural education framework, GenAI tools can support students in interpreting site and socio-cultural conditions and in structuring early-stage design reasoning, especially when context awareness is treated as a design requirement rather than assumed by the model [9,64]. During the pre-design phase, AI-assisted workflows can help integrate spatial and environmental analyses to support more informed early decisions and design communication [47]. However, the educational value of such applications depends on embedding these tools within pedagogical approaches that emphasize critical interpretation and responsible use, given the risks of unreliable outputs and the need for human-centered oversight and risk awareness [69]. Accordingly, GenAI should be positioned as a supportive analytical aid rather than a substitute for students' contextual understanding and design judgment [77]. Meron and Tekmen Araci [43] conducted a semi-structured exploratory conversation with ChatGPT to investigate its potential as a virtual colleague. Their study consisted of multiple case studies; the first one involved working alongside ChatGPT to prepare a design student assignment, while the second one involved utilizing ChatGPT to develop a postgraduate design studio course. They found that ChatGPT accelerated the creation of structured templates for course development and proved to be a useful tool for brainstorming and ideation, which improves time management and reduces workload. Building on this, Çınar Kalenderoğlu and Demiröz [20] highlighted that developing precise text prompts is crucial in early design phases, helping students better control AI outputs and align them with project objectives. Additionally, Cheung et al. [9] extended this idea by proposing a conversational AI-enhanced framework for the early-stage performative design process, showing how a multimodal agentic AI bot can assist students during site analysis, massing optimization, and context-based design exploration. By integrating text and image processing within a group chat environment, students were able to negotiate design ideas, test different massing options, and receive feedback that supports iterative decision-making. Moreover, Al-Soufi and El Shafie [6] showed that AI tools like ArchiteChures can support the pre-design phase by generating initial floor plan layouts based on input parameters and site constraints, which are then iteratively refined to align with local regulations and project needs.

Braiden et al. [8] confirmed through a professional survey that AI is mainly applied in early design phases for proposals, site data collection, and initial concept generation with AI-based imagery, while Belaroussi and Martín-Gutierrez [7] illustrated how GPT-4V can support this stage by analyzing urban ambiance and spatial mood to inform better contextual understanding before conceptual development. Similarly, Cao et al. [19] highlighted that artificial intelligence proves particularly suited to conceptualization and narrative-based design tasks, supporting early-stage design thinking and analysis. In addition, Rodriguez et al. [14] demonstrated that combining AI-based image generation with Virtual Reality

helps architects analyze the perception and sensations of virtual spaces early on, offering immersive insight into spatial qualities before advancing to detailed development.

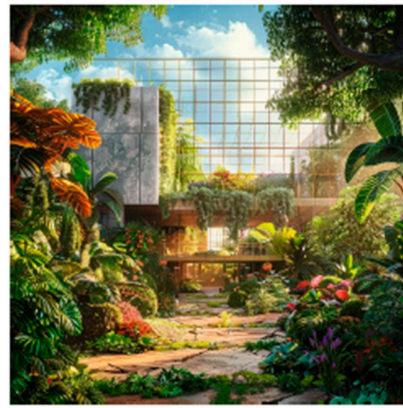
However, ChatGPT showed a limited ability to fully replace humans and required guidance from experienced educators in complex and context-specific tasks. Despite the benefits, AI chatbots may struggle with higher-order thinking and may provide misleading data. Thus, it is crucial to review results while using them. This was also confirmed by Caliskan [39], who conducted an interview with ChatGPT in January 2023 to explore its potential in shaping third-year architectural design studio work. The result showed that while ChatGPT offers valuable suggestions and generates a broad range of ideas for design studio projects, a collaborative approach, combining instructor experience, education, and various research resources, is indispensable.

3.3.2. Conceptual Design

This is the next stage of the design process, where the conceptual framework of a project is developed. During this stage, designers explore multiple design concepts using a variety of methods and tools, including sketching. Gen AI tools can potentially generate and visualize design concepts effectively, as demonstrated in several studies. This has been reported to influence the conceptual design process in architecture by offering advanced tools that expand the boundaries of creativity and exploration [50]. A key potential contribution of AI is its ability to generate a wide range of unique, out-of-the-box design concept ideas from text-based inputs. AI-based platforms, for example, may retrieve images and designs from a massive database, providing architects with a range of visual stimuli to spark new ideas and offer unexpected solutions. Also, Lekesiz and Müezzinoğlu [13] demonstrated how text-to-image AI tools in a third-year design studio helped students expand their conceptual thinking, produce diverse visual outputs, and translate abstract ideas into spatial representations.

Karadağ and Ozar [12] highlighted how text-to-image generators like Midjourney can broaden students' design possibilities by translating textual prompts into rich visual material. Their study demonstrated that utilizing T2I AI in a design studio setting enables students to explore multiple design directions, experiment with abstract ideas, and refine conceptual alternatives through high-fidelity, quickly generated visuals. Schroth and Maier [15] also illustrated how text-to-image prompts can be used in conceptual design stages to translate descriptive textual inputs into detailed visual outputs, helping students and designers rapidly test variations and expand creative exploration, as shown in Figure 3. This iterative process not only stimulates creative thinking but also helps bridge the gap between abstract concepts and tangible spatial representations. Paananen et al. [33] similarly emphasized that text-to-image generation supports serendipitous discovery and an imaginative mindset during the early design stage, enriching the ideation process with unexpected visual outcomes. However, Deregibus [10] argued that while this keyword-based approach can produce impressive analogical results, it often recycles existing visual patterns, reinforcing organized stereotypes instead of producing genuinely revolutionary content.

In line with this, Maksoud et al. [31] demonstrated how Midjourney was applied to support the creative brainstorming phase by generating conceptual forms of a Safavid mosque, highlighting how combining text and image prompts can help produce rapid, diverse design variations that align with historical architectural qualities. Similarly, Jin et al. [25] demonstrated how GAN models can integrate traditional architectural elements into new conceptual designs, enabling designers to automatically generate façade compositions and layouts by learning from large datasets of historical components. This approach enriches conceptual exploration and supports the development of culturally embedded design alternatives.



Prompt:
/imagine – create a photo of a futuristic botanical garden with a building in the foreground and in the golden ratio and a lot of colorful plants in the background. Natural green colors, sunny, high detail, photorealistic

Figure 3. Example of a text-to-image AI prompt in Midjourney, with detailed descriptive input. Source: Schroth and Maier [15], used under Creative Commons Attribution license CC BY-ND 4.0.

Zwangsleitner et al. [37] illustrated how combining ChatGPT and Midjourney in an architecture seminar expanded students' conceptual thinking by turning text prompts into sketches and AI-generated visual variations. This iterative loop between analog and AI tools helped students test multiple alternatives and is illustrated in Figure 4, which shows examples of student work combining sketching and AI-rendered outputs.

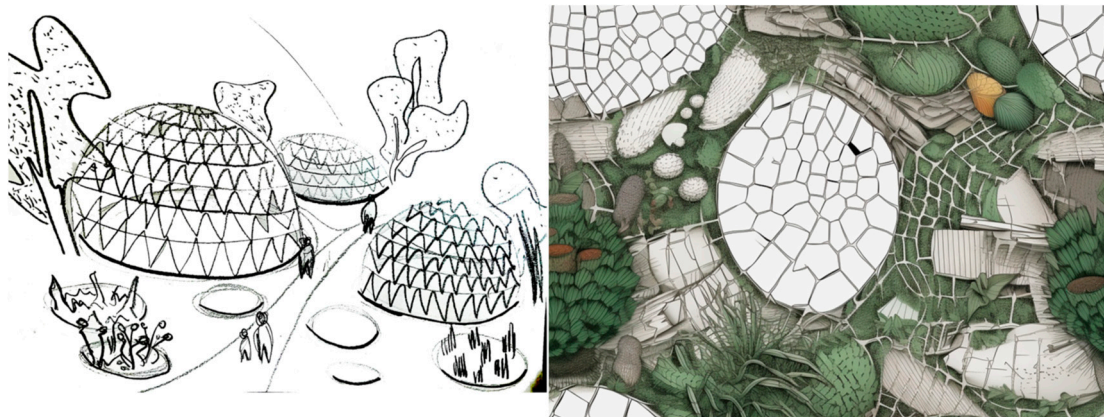


Figure 4. Conceptual sketches generated with ChatGPT and Midjourney. Source: Zwangsleitner et al. [37], used under Creative Commons Attribution license CC BY-ND 4.0.

GenAI tools could also help students in image generation and rapid concept development [21,22]. Asfour [17] examined AI's role in various stages of the design process from an academic standpoint, including the conceptual design stage. The study presented sample student work on the use of physical models and Midjourney application to generate architectural compositions.

GenAI tools could also be effectively used to explore architectural forms inspired by nature. By incorporating these forms into the design exploration process, architects can explore a variety of creative ideas that enhance their design [42,47]. Derevyanko and Zalevska [40] compared the use of Midjourney, Stable Diffusion, and DALL-E in this regard. The outcome showed that Midjourney and DALL-E could generate creative ideas for students and provide expressive visuals, while Stable Diffusion helps students create visualizations for their projects. Similarly, Ploennigs and Berger [49] compared three common GenAI tools, namely Midjourney, DALL-E, and StableDiffusion, using case studies. The results demonstrated how these tools could automate several design tasks, including concept generation. These tools can be classified as text-to-image AI tools, where users can generate images using text prompts that are processed using AI language models. The use of parametric design also allows students to

define a set of design parameters to create several conceptual architectural design alternatives using several tools such as Archistar, Conix, and Spacemaker.

3.3.3. Design Development

At this phase, a design model is developed to satisfy the functional and aesthetic requirements, and detailed drawings are created for further refinement. AI tools could effectively help to fulfill these tasks with more efficiency and creativity. In the reviewed development-stage literature, the evidence is predominantly technical/professional [30,48], and explicit learning outcomes are less frequently reported. Accordingly, this stage often involves Grasshopper and Dynamo, which are visual programming tools that are commonly used. However, Dynamo offers more advantages to enhance collaboration with a broader range of stakeholders, particularly within BIM-based workflows [38]. Parametric design tools, such as Rhinoceros 3D combined with Grasshopper, are used for creating adaptive and complex models, transforming the design process by enabling detailed simulations and automating tasks. Sketch-to-image, or sketch-to-architectural technique, is another powerful AI-assisted technique in design development stage, where designers could draw the layout, and an AI tool could be used to convert it to an image. Li et al. [30] utilized GenAI to convert simple sketches into conceptual floor plans and 3D models. They used the Stable Diffusion tool for image generation and integrated both Low-Rank Adaptation (LoRA) and ControlNet models to improve the outputs. The generated images were then processed using Rhino and Grasshopper to create rendered 3D models. This approach enables precise control over the input text, accelerates the early design phases, and provides creative possibilities for architects. Figure 5 illustrates a clear workflow of AI-generated architectural design, showing how sketches and floor plans are processed through GenAI tools to produce depth maps, 3D models, and stylistic variations. This visual workflow demonstrates how a conceptual sketch can evolve into a detailed and diverse architectural design through iterative AI processing and editing [30].

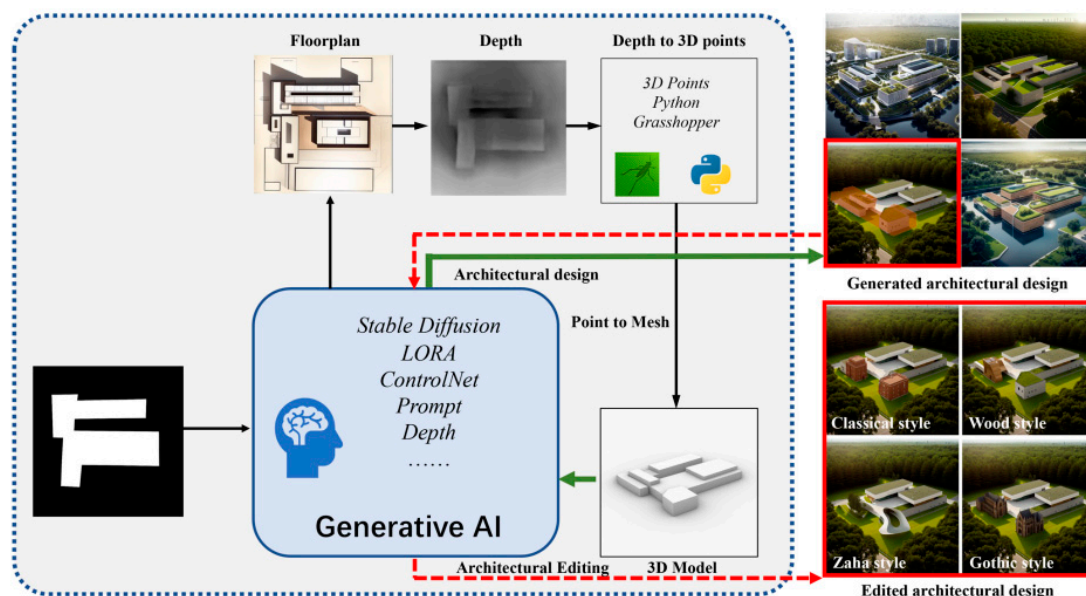


Figure 5. AI workflow for transforming sketches into 3D architectural designs with style variations. Source: Li et al. [30], used under Creative Commons Attribution license CC BY 4.0.

Moreover, AI tools may be used to improve energy efficiency and enhance sustainability while developing building design. GenAI tools, such as GANs and CA, help architects design buildings that balance aesthetics and energy efficiency by simulating environmental factors [44]. These models are further supported by AI-driven systems that integrate renew-

able energy sources and optimize systems like heating, ventilation, and air conditioning (HVAC) and lighting for long-term sustainability [27]. Khogali [29] highlighted that AI-powered design solutions allow architects to generate green building designs incorporating features such as green roofs, terraces, and natural lighting, significantly reducing energy use and ensuring long-term sustainability. These features and AI-driven optimization models may reduce energy use by up to 20%, further advancing sustainable architecture [28].

3.3.4. Design Production

The design production phase involves transforming developed design models into refined design outputs that are prepared to communicate the project. Platforms such as Lookx.AI, PromeAI, Minimal.ai, and Rerender.ai can enable users to produce realistic 3D renderings with advanced control over materials, lighting, and environmental context. These tools utilize ML algorithms to automate rendering techniques that are traditionally handled manually through software like Lumion or V-Ray. Cudzik and Nyka [21] highlighted how AI image-generation tools enhance the design production phase by enabling students to quickly produce high-quality visuals that support design communication. The use of GenAI reduced the time spent on traditional rendering, allowing for a greater focus on refining the conceptual and expressive aspects of design. Khogali [29] emphasized the significant role of AI in 3D rendering and architectural visualization. For example, as illustrated in Figure 6, AI can bring the initial architectural design to life through detailed renderings that integrate sustainability principles, such as green roofs, terraces, and natural lighting, demonstrating how GenAI tools contribute to producing expressive and environmentally responsive visualizations.

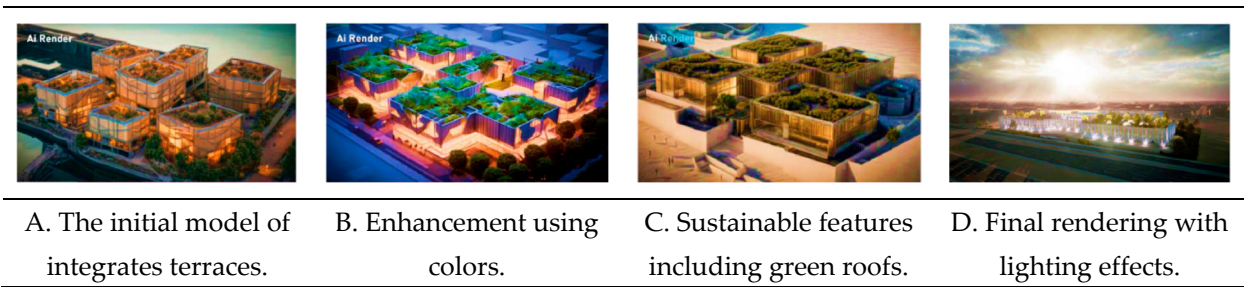


Figure 6. GenAI use to produce expressive and environmentally responsive visualizations. Source: Khogali [29], used under Creative Commons Attribution license CC BY 4.0.

Jo et al. [26] illustrated how GenAI models can generate contextually relevant façade imagery in the design production stage, bridging the gap between conceptual proposals and photorealistic representations. By producing detailed façade images and urban perspectives at an early stage, students and designers can iterate design options faster, improve visual storytelling, and align final outputs with contextual and cultural cues while reducing production time. This demonstrates how GenAI can bridge the gap between conceptual design outputs and polished presentation materials, enhancing both workflow efficiency and visual quality in the design production stage. GenAI tools could generate realistic 3D models that help designers better present their designs and quickly produce renderings with realistic lighting, textures, and materials, improving the design communication process [23]. Fernberg et al. [41] also demonstrated how AI-powered image generators can support the design production stage by creating high-quality, customizable 2D asset libraries, which significantly reduce the time spent on entourage elements and allow designers to maintain creative control while streamlining repetitive tasks.

Günaydın et al. [24] highlighted that AI co-authoring with designers allows the transformation of textual ideas into visual outputs, streamlining the creation of high-quality

design representations. El Moussaoui [11] noted that many AI tools extend beyond early design to support the production of construction documents, integrating representation and data analytics to finalize architectural outputs. Similarly for urban design, Shokry [16] concluded that AI tools such as Test Fit, Plan Finder, and Luma can significantly enhance sustainability by optimizing land use, streamlining layout planning, and helping stakeholders better understand the benefits of sustainable design through realistic visualizations.

Furthermore, in the design production stage, construction materials and technologies should be clearly defined. Rane [45] highlighted the role of AI in this regard, enabling architects to make informed, data-driven decisions. Baduge et al. [48] focused on AI's predictive capabilities in material performance. Using ML and DL models, AI forecasts the durability and strength of materials like concrete and steel, ensuring they meet performance standards, reducing waste and enhancing efficiency and sustainability in construction. Informed selection of construction materials is usually guided by life cycle assessment (LCA). Bassey et al. [18] mentioned that AI-driven LCA models use large datasets to enhance the accuracy and depth of environmental assessments. These models can adapt to new data, allowing for real-time monitoring and dynamic adjustments, which help ensure continuous sustainability throughout a building's lifecycle. Furthermore, Płoszaj-Mazurek and Ryńska [34] highlighted that integrating AI with ML and BIM allows for more accurate carbon footprint estimates, design optimizations, and improved decision-making in architectural projects. However, to ensure practicality, GenAI-generated design outputs should be treated as preliminary and require verification and refinement to comply with country-specific building codes and regulatory requirements [45]. In educational contexts, Tabrizi et al. [46] argued that AI simplifies complex LCA processes for students by automating data collection, predictive modeling, and visualization. Overall, later-stage applications are more frequently reported as professional workflows; therefore, educational implications are discussed only when studies explicitly report learning-related evidence or are clearly transferable to studio teaching. Table 3 provides a comparative synthesis across design stages, highlighting evidence type and reported learning outcomes where available.

Table 3. Stage-level synthesis across the architectural design stages.

Design Stage	Dominant Tools	Evidence Type	Reported Educational Outcomes
Pre-design	LLM chatbots (e.g., ChatGPT and comparable tools), multimodal agents; AI + VR; layout generators	Mostly empirical (studio/workshops, surveys/interviews)	Faster info gathering; prompt literacy; early decision support; context-aware analysis
Conceptual	Text-to-image generators (Midjourney/SD/DALL-E); prompting; hybrid analog + AI; parametric exploration tools	Mixed, empirical + technical comparisons + conceptual critiques	Expanded ideation; rapid visualization; creative exploration; risk of pattern recycling
Development	Parametric/Computational workflows (Rhino-Grasshopper, Dynamo); sketch-to-architecture pipelines (e.g., Stable Diffusion + Rhino/Grasshopper); performance/sustainability optimization models	Mostly technical/professional	Workflow acceleration; performance/sustainability support (learning outcomes are less reported)
Production	AI rendering/visualization platforms (LookX.AI, PromeAI, etc.); façade generation; asset libraries; ML + BIM + LCA tools	Mixed (educational + technical + practice-oriented)	Faster high-quality outputs; improved communication; reduced repetitive tasks; LCA learning support (often practice-oriented; requires code compliance verification)

4. Discussion

Based on the findings presented in this study, it is widely accepted that using GenAI in architectural design courses can support cognitive offloading by automating time-consuming routine tasks, thereby allocating more time for creative work and critical reflection. Students who utilize AI in the architectural design process are more likely to produce more creative and innovative design solutions than those who design using traditional design methods [21,29]. It is also suggested that students who use GenAI in their design projects are more likely to experience improved time management and reduced anxiety, allowing them to explore multiple design alternatives without a repetitive workload. When embedded within studio teaching, this potential may extend beyond offloading toward cognitive amplification by expanding exploration and iteration, rather than only accelerating production [51]. However, these benefits depend on pedagogy: without clear learning objectives and assessment criteria, GenAI may shift studio work toward outcome-driven design (polished outputs) at the expense of process learning (reasoning, iteration, and critique), particularly under over-reliance [42]. This reinforces the need for educator readiness and consistent guidance to integrate GenAI responsibly and ethically, especially given the rapid pace of tool development [5].

In studio-based learning, architectural design development is commonly framed as an iterative social cycle of proposal, critique, and refinement, where students progressively build judgment through repeated evaluation of their decisions [59,63]. When GenAI is introduced, this cycle can be extended pedagogically by making instructor and peer critique target not only the generated artifact but also the student's prompting, selection, and editing strategies as part of process learning [20,24]. This interactive "human feedback loop" can be framed as a pedagogical metaphor through the lens of Reinforcement Learning with Human Feedback (RLHF). RLHF aligns AI model behavior by iteratively incorporating human preference signals that shape subsequent outputs [78]. Similarly, studio critique can function as the preference signal that guides the student's next "actions", refining prompts, curating alternatives, and justifying critical edits, so GenAI becomes a structured environment for developmental learning rather than a shortcut to polished outcomes. Accordingly, assessment can focus on student control (agency), e.g., prompt rationale, curation decisions, and critical edits, rather than only final output quality [24].

Despite the benefits of adopting AI into architectural education, it also poses several risks and challenges [77]. From an ethical perspective, GenAI algorithms rely on large datasets that may embed cultural, social, and contextual biases. When applied uncritically in architectural education, these systems may reinforce homogenized design solutions, overlook local identity, and marginalize community-specific needs. Furthermore, AI-driven search and content-generation mechanisms may influence students' design thinking by prioritizing patterns derived from existing data rather than fostering original, context-sensitive concepts. This raises ethical concerns related to data privacy, algorithmic bias, and the potential regeneration of cultural stereotypes, particularly when GenAI tools are trained on imbalanced datasets [79,80]. In addition, the over-reliance on AI tools may weaken key manual design skills and critical thinking abilities [42]. The high costs associated with GenAI use in architectural education are another concern. Institutions with limited resources may not be able to afford the necessary software, hardware, and training. To address these challenges, GenAI integration into architectural design education must be managed carefully to balance the technical advancements with the need to satisfy the core design skills. This necessitates that educators ensure AI tools are used to complement human creativity and critical thinking rather than replace them [81,82]. Educators now face the responsibility of teaching students how to use GenAI ethically and thoughtfully, augmenting human creativity within studio learning. [5].

In line with broader AI in architectural pedagogy that highlights both opportunities and risks of generative systems in learning contexts, architecture-focused debates similarly emphasize that GenAI should augment, rather than replace, human judgment and studio learning practices [77,81,82]. Few studies, however, have systematically addressed how GenAI could be consciously integrated into the different design stages, including pre-design analysis, conceptual design, design development, and design refinement and production. While conceptual applications of GenAI have been widely explored in the literature review presented in this study, their impact on the advanced design phases remains underexplored in research. The long-term impact of this integration is still unclear. Regardless of differing views, it is suggested that AI is expected soon to become an integral component of architectural practice and education. Thus, there is an urgent need to investigate how it could affect our educational systems and design practices. Accreditation bodies should also update their standards to guide architecture schools in adapting to these changes [17]. To bridge these gaps, future research should focus on:

- Evaluating the long-term influence of GenAI on architectural education, including its effects on design thinking, creativity, and students' engagement in the market.
- Exploring the integration of GenAI across all stages of the architectural design process, particularly within the design development and refinement stages.
- Examining best practices for pedagogy development that combine AI utilization with traditional design methods.
- Developing ethical guidelines and educational frameworks that address the responsible use of AI-generated content.
- Improve the readiness of academic institutions for AI utilization through professional development programs and policy support.

5. Conclusions

Architectural design education plays a crucial role in preparing architects capable of addressing complex environmental, social, and technological challenges. Traditionally, it has emphasized creativity, critical thinking, and problem-solving skills essential for translating conceptual ideas into spatial, functional, and sustainable design solutions. Architectural design education has undergone significant transformation in recent years, with innovative teaching methods being developed to take advantage of the disruptive design technologies, including GenAI. This review aimed to explore the current applications of GenAI in architectural design education and the associated opportunities and challenges in this regard. It considered the different design stages, namely the pre-design analysis, concept generation, design development, and design production. The study found that GenAI tools are increasingly influencing design tasks and creative processes to support ideation, visualization, and design iteration. This necessitates an in-depth evaluation of the traditional architectural design conventions commonly used in the design studio and their relevance to the needed skills nowadays in the market. Through enabling data-driven decision-making, students' creativity could be enhanced by enabling innovative design concept development, advanced visualization techniques, and interactive learning experiences.

The reviewed literature, in general, highlighted the advantages of GenAI utilization in architectural design education to enhance innovation and support data-driven decision-making using a variety of tools. This streamlines design workflow and enhances students' productivity throughout the various stages of the architectural design process, especially in the early design phases, which are highlighted in the literature. In the pre-design analysis stage, GenAI tools assist students in collecting design data, generating initial design ideas, and analyzing contextual factors, thereby supporting more informed and responsive design strategies. In the conceptual design stage, GenAI platforms enable students to visualize and

evaluate multiple conceptual design alternatives. More research is needed to address the role of AI in the advanced design stages, such as design refinement. Despite the increasing enthusiasm for incorporating GenAI into architectural design education, significant gaps and challenges remain. A key issue is the absence of empirical research on the long-term effects of GenAI on students' creativity, critical thinking, and problem-solving abilities. While existing studies often highlight GenAI's potential to foster innovation and improve efficiency, few have investigated its long-term impact on the educational experience of architecture students.

There is also a need to address the risk of over-reliance on AI tools and the weakening of manual and critical design skills, in addition to some ethical concerns such as data bias, cultural misrepresentation, and intellectual property rights. This resulted in different institutional pedagogical approaches in this regard, with different levels of AI integration into architectural design education. Future research should also examine the feasibility of complex GenAI integration approaches in architectural education, particularly in terms of cost, infrastructure, and institutional capacity. Incorporating process flowcharts and clearly defined software workflows would enhance contextual clarity and provide more actionable guidance for implementation. There is also a need to investigate the new skills needed in the market and re-design academic programs to respond to them. This requires making graduates more digitally enabled, not only in the design-related skills but also in other domains such as construction automation and smart building management. The study also recommends developing best practice guidelines and quality assurance systems to guide the ethical use of GenAI in architectural design education and determine the "best mix" of traditional and AI-based design methods across the different design phases.

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Abbreviations

The following abbreviations are used in this manuscript:

ADE	Automatic Database Exclusion
AI	Artificial Intelligence
ANN	Artificial Neural Networks
BIM	Building Information Modeling
DL	Deep Learning
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing

References

- Chęć-Małyszek, A. Architecture as the art of creating human-friendly places, Lublin public space. *Bud. I Archit.* **2021**, *20*, 59–76. [CrossRef]
- Canestrino, G. Considerations on optimization as an architectural design tool. *Nexus Netw. J.* **2021**, *23*, 919–931. [CrossRef]
- Ibrahim, N.L.N.; Utaberta, N. Learning in architecture design studio. *Procedia-Soc. Behav. Sci.* **2012**, *60*, 30–35. [CrossRef]
- Shi, Y.; Xu, C.; Xu, Y. Research on the practice of the new education mode reform based on the teaching of architectural design course and the thinking of future architectural education. *Nanotechnol. Environ. Eng.* **2021**, *6*, 49. [CrossRef]
- Desouki, M.; Shiba, A.S.E.; Elboshy, B. An effective architectural criticism teaching framework link and enhance design skills. *Adv. Eng. Sci.* **2022**, *54*, 1595–1605. [CrossRef]
- Al-Soufi, N.; El Shafie, H. Assessment of the Current State of AI-Aided Design in Architecture Case Study: Design of a Residential Building in Riyadh City. *J. Al-Azhar Univ. Eng. Sect.* **2025**, *20*, 259–314. [CrossRef]
- Belaroussi, R.; Martín-Gutierrez, J. Architectural ambiance: ChatGPT versus human perception. *Electronics* **2025**, *14*, 2184. [CrossRef]
- Braiden, H.; Chamberlain, B.; George, B.H.; Fernberg, P. AI in Practice: Professional Survey Findings from Landscape Architects in North America. *J. Digit. Landsc. Archit.* **2025**, *2025*, 626–633. [CrossRef]
- Cheung, L.H.; Wang, L.; Lei, D. Conversational, agentic AI-enhanced architectural design process: Three approaches to multi-modal AI-enhanced early-stage performative design exploration. *Archit. Intell.* **2025**, *4*, 10. [CrossRef]
- Deregibus, C. The Key Role of Keywords in Architectural Design: A Systemic Framework. *Architecture* **2025**, *5*, 32. [CrossRef]
- El Moussaoui, M. Architectural practice process and artificial intelligence—An evolving practice. *Open Eng.* **2025**, *15*, 20240098. [CrossRef]
- Karadağ, D.; Ozar, B. A new frontier in design studio: AI and human collaboration in conceptual design. *Front. Archit. Res.* **2025**, *14*, 1536–1550. [CrossRef]
- Lekesiz, G.; Müezzinoğlu, C. An approach to AI-supported learning in architectural education: Case of speculative space design. *A/Z ITU J. Fac. Archit.* **2025**, *22*, 217–231. [CrossRef]
- Rodriguez, J.K.C.; Yali, J.B.A.; Torres, V.S.M. Technology and Architecture: Impact of Artificial Intelligence and Virtual Reality on the Perception of Architectural Design. *Civ. Eng. Archit.* **2025**, *13*, 637–652. [CrossRef]
- Schroth, O.; Maier, A. Integrating Generative Artificial Intelligence into the Landscape Architecture Design Process. *J. Digit. Landsc. Archit.* **2025**, *2025*, 665–675. [CrossRef]
- Shokry, R. Exploring the Impact of Artificial Intelligence on Urban Design. *JES J. Eng. Sci.* **2025**, *53*, 457–473. [CrossRef]
- Asfour, O.S. How artificial intelligence could affect the future of architectural design education. In Proceedings of the 8th International Conference on Green Urbanism, Rome, Italy, 7–9 October 2024. [CrossRef]
- Bassey, K.E.; Juliet, A.R.; Stephen, P.O. AI-Enhanced lifecycle assessment of renewable energy systems. *Eng. Sci. Technol. J.* **2024**, *5*, 2082–2099. [CrossRef]
- Cao, Y.; Gao, X.; Yin, H.; Yu, K.; Zhou, D. Reimagining tradition: A comparative study of artificial intelligence and virtual reality in sustainable architecture education. *Sustainability* **2024**, *16*, 11135. [CrossRef]
- Çınar Kalenderoğlu, S.; Demiröz, M. Integrating Text-to-Image AI in Architectural Design Education: Analytical Perspectives from a Studio Experience. *J. Des. Studio* **2024**, *6*, 247–258. [CrossRef]
- Cudzik, J.; Nyka, L. Artificial intelligence in architectural education—Green campus development research. *Glob. J. Eng. Educ.* **2024**, *26*, 20–25. Available online: <https://www.wiete.com.au/journals/GJEE/Publish/vol26no1/03-Cudzik-J.pdf> (accessed on 15 December 2025).
- Fareed, M.W.; Bou Nassif, A.; Nofal, E. Exploring the potentials of artificial intelligence image generators for educating the history of architecture. *Heritage* **2024**, *7*, 1727–1753. [CrossRef]
- Golkarian, S. Enhancing architectural space through AI-driven ideation: A case study of future Iranian-traditional city. *Rev. Amazon. Investig.* **2024**, *13*, 157–172. [CrossRef]
- Günaydın, C.; Kasalı, A.; Doğan, F. Artificial Intelligence as a Pedagogical Tool for Architectural Design Education. *J. Des. Studio* **2024**, *6*, 219–229. [CrossRef]
- Jin, D.; Zairul, M.; Salih, S.A. Computational analysis of traditional architectural elements impact on AI-generated designs. *Oper. Res. Eng. Sci. Theory Appl.* **2024**, *7*, 176–196. [CrossRef]
- Jo, H.; Lee, J.-K.; Lee, Y.-C.; Choo, S. Generative artificial intelligence and building design: Early photorealistic render visualization of façades using local identity-trained models. *J. Comput. Des. Eng.* **2024**, *11*, 85–105. [CrossRef]
- Karadag, I.; Yıldız, A. Artificial intelligence in architecture: Innovations, challenges, and ethical considerations. In *Making Art with Generative AI Tools*; Hai-Jew, S., Ed.; IGI Global Scientific Publishing: Hershey, PA, USA, 2024; pp. 215–225. [CrossRef]
- Karimi, H.; Adibhesami, M.A.; Hoseinzadeh, S.; Salehi, A.; Groppi, D.; Astiaso Garcia, D. Harnessing deep learning and reinforcement learning synergy as a form of strategic energy optimization in architectural design: A case study in Famagusta, North Cyprus. *Buildings* **2024**, *14*, 1342. [CrossRef]

29. Khogali, H.A. The effect of applying artificial intelligence in architecture college developing design process. *J. Sustain. Dev.* **2024**, *17*, 35–49. [\[CrossRef\]](#)
30. Li, P.; Li, B.; Li, Z. Sketch-to-architecture: Generative AI-aided architectural design. In *Proceedings of the Pacific Graphics 2023, Daejeon, Republic of Korea, 10–13 October 2023*; The European Association for Computer Graphics: Geneva, Switzerland, 2024.
31. Maksoud, A.; Elshabshiri, A.; Saeed Hilal Humaid Alzaabi, A.; Hussien, A. Integrating an Image-Generative Tool on Creative Design Brainstorming Process of a Safavid Mosque Architecture Conceptual Form. *Buildings* **2024**, *14*, 843. [\[CrossRef\]](#)
32. Montenegro, N. Integrative analysis of Text-to-Image AI systems in architectural design education: Pedagogical innovations and creative design implications. *J. Archit. Urban.* **2024**, *48*, 109–124. [\[CrossRef\]](#)
33. Paananen, V.; Oppenlaender, J.; Visuri, A. Using text-to-image generation for architectural design ideation. *Int. J. Archit. Comput.* **2024**, *22*, 458–474. [\[CrossRef\]](#)
34. Płoszaj-Mazurek, M.; Ryńska, E. Artificial intelligence and digital tools for assisting low-carbon architectural design: Merging the use of Machine Learning, Large Language Models, and Building Information Modeling for Life Cycle Assessment tool development. *Energies* **2024**, *17*, 2997. [\[CrossRef\]](#)
35. Sindhu Devi, A.; Maruthuperumal, S. Introduction to ChatGPT. *Int. J. Sci. Res.* **2024**, *13*, 108–109. [\[CrossRef\]](#)
36. Xu, H.; Omitaomu, F.; Sabri, S.; Zlatanova, S.; Li, X.; Song, Y. Leveraging generative AI for urban digital twins: A scoping review on the autonomous generation of urban data, scenarios, designs, and 3D city models for smart city advancement. *Urban Inform.* **2024**, *3*, 29. [\[CrossRef\]](#)
37. Zwangsleitner, F.; Habjanič, G.; Knegetendorf, A. AI as a Tool in the Landscape Architecture Design Process. *J. Digit. Landsc. Archit.* **2024**, *2024*, 987–994. [\[CrossRef\]](#)
38. Bölek, B.; Tural, O.; Özbaşaran, H. A systematic review on artificial intelligence applications in architecture. *J. Des. Resil. Archit. Plan.* **2023**, *4*, 91–104. [\[CrossRef\]](#)
39. Caliskan, E.B. Interview with Chat GPT to define architectural design studio work: Possibilities, conflicts and limits. *J. Des. Studio* **2023**, *5*, 57–71. [\[CrossRef\]](#)
40. Derevyanko, N.; Zalevska, O. Comparative analysis of neural networks Midjourney, Stable Diffusion, and DALL-E and ways of their implementation in the educational process of students of design specialties. *Sci. Bull. Mukachevo State Univ. Ser. Pedagog. Psychol.* **2023**, *9*, 36–44. [\[CrossRef\]](#)
41. Fernberg, P.; George, B.H.; Chamberlain, B. Producing 2D Asset Libraries with AI-powered Image Generators. *J. Digit. Landsc. Archit.* **2023**, *2023*, 186–194. [\[CrossRef\]](#)
42. Desouki, M.; ElHaddad, T.A.; ElBoshey, B. Revolutionary artificial intelligence architectural design solutions; is it an opportunity or a threat? *Mansoura Eng. J.* **2023**, *48*, 11. [\[CrossRef\]](#)
43. Meron, Y.; Tekmen Araci, Y. Artificial intelligence in design education: Evaluating ChatGPT as a virtual colleague for post-graduate course development. *Des. Sci.* **2023**, *9*, 30. [\[CrossRef\]](#)
44. Milošević, J.; Đukanović, L.; Živković, M.; Žujović, M.; Gavrilović, M. Automated compositions: Artificial intelligence aided conceptual design explorations in architecture. In *Proceedings of the 9th International Scientific Conference on Geometry and Graphics MONGEOMETRIJA 2023, Novi Sad, Serbia, 7–10 June 2023*; Serbian Society for Geometry and Graphics SUGIG: Belgrade, Serbia, 2023; pp. 103–115. Available online: https://raf.arh.bg.ac.rs/bitstream/handle/123456789/1282/bitstream_4420.pdf?sequence=1&isAllowed=y (accessed on 16 January 2026).
45. Rane, N. Potential role and challenges of ChatGPT and similar generative artificial intelligence in architectural engineering. *Int. J. Artif. Intell. Mach. Learn.* **2023**, *4*, 22–47. [\[CrossRef\]](#)
46. Tabrizi, T.B.; Gocer, O.; Sadrieh, A.; Globa, A. Leveraging AI to instruct architecture students on circular design techniques and life cycle assessment. In *Proceedings of the 9th International Conference on Higher Education Advances, Valencia, Spain, 19–22 June 2023*; Editorial Universitat Politècnica de València: València, Spain, 2023; pp. 647–655. [\[CrossRef\]](#)
47. Yudhanta, W.C.; Hadinata, I.Y. Computational methods and artificial intelligence in the architectural pre-design process (case study: House design). *J. Artif. Intell. Archit.* **2023**, *2*, 53–60. [\[CrossRef\]](#)
48. Baduge, S.K.; Thilakarathna, S.; Perera, J.S.; Arashpour, M.; Sharafi, P.; Teodosio, B.; Shringi, A.; Mendis, P. Artificial intelligence and smart vision for building and construction 4.0: Machine and deep learning methods and applications. *Autom. Constr.* **2022**, *141*, 104440. [\[CrossRef\]](#)
49. Ploennigs, J.; Berger, M. AI art in architecture. *AI Civ. Eng.* **2022**, *2*, 8. [\[CrossRef\]](#)
50. Castro Pena, M.L.; Carballal, A.; Rodríguez-Fernández, N.; Santos, I.; Romero, J. Artificial intelligence applied to conceptual design. A review of its use in architecture. *Autom. Constr.* **2021**, *124*, 103550. [\[CrossRef\]](#)
51. Kee, T.; Kuys, B.; King, R. Generative artificial intelligence to enhance architecture education to develop digital literacy and holistic competency. *J. Artif. Intelligence Archit.* **2024**, *3*, 24–41. [\[CrossRef\]](#)
52. Ayman, A.; Mansour, Y.; Eldaly, H. Generative vs. non-generative AI: Analyzing the effects of AI on the architectural design process. *Eng. Res. J.* **2024**, *53*, 119–128. [\[CrossRef\]](#)

53. Yan, L.; Greiff, S.; Teuber, Z.; Gašević, D. Promises and challenges of generative artificial intelligence for human learning. *Nat. Hum. Behav.* **2024**, *8*, 1839–1850. [CrossRef] [PubMed]
54. Salama, A.M. *Spatial Design Education: New Directions for Pedagogy in Architecture and Beyond*, 1st ed.; Routledge: Oxfordshire, UK, 2015. [CrossRef]
55. Afify, H.M.N. *From Beginning to End: The Design Process*; College of Architecture and Planning, University of Dammam: Dammam, Saudi Arabia, 2014.
56. Bozkurt, E. Integration of theory courses and design studio in architectural education using sustainable development. *SHS Web Conf.* **2016**, *26*, 01102. [CrossRef]
57. AboWardah, E.S. Bridging the gap between research and schematic design phases in teaching architectural graduation projects. *Front. Archit. Res.* **2020**, *9*, 82–105. [CrossRef]
58. Soliman, A. Appropriate teaching and learning strategies for the architectural design process in pedagogic design studios. *Front. Archit. Res.* **2017**, *6*, 204–217. [CrossRef]
59. Masatlıoğlu, C.S.E.; Balaban, Ö.C. Reflective thinking and self-assessment: A model for the architectural design studio. *J. Des. Resil. Archit. Plan.* **2024**, *5*, 35–49. [CrossRef]
60. Milošević, J. Implementing a blended design studio model in architectural engineering education. *Int. J. Eng. Educ.* **2021**, *37*, 1–13. Available online: <https://raf.arh.bg.ac.rs/handle/123456789/2496> (accessed on 16 January 2026).
61. Badawi, A.M.; Abdullah, M.R. Interdisciplinary design education: Development of an elective course in architecture and engineering departments. *J. Eng. Appl. Sci.* **2021**, *68*, 10. [CrossRef]
62. Dervisoglu, C.D.; Yilmaz, E. Examining the effect of learning environment on student behaviour through comparison of face-to-face and online design studio. *MEGARON/Yıldız Tech. Univ. Fac. Archit. E-J.* **2023**, *18*, 142–157. [CrossRef]
63. Grover, R.; Wright, A. Shutting the studio: The impact of the COVID-19 pandemic on architectural education in the United Kingdom. *Int. J. Technol. Des. Educ.* **2023**, *33*, 1173–1197. [CrossRef]
64. Birgonul, Z.; Carrasco, O. The multidimensional exploration methodology for architectural design studio education: An innovative teaching methodology proposal. *SPACE Int. J. Conf. Proc.* **2022**, *2*, 11–21. [CrossRef]
65. Mahmoud Saleh, M.; Abdelkader, M.; Sadek Hosny, S. Architectural education challenges and opportunities in a post-pandemic digital age. *Ain Shams Eng. J.* **2023**, *14*, 102027. [CrossRef]
66. Tripathi, S. Artificial intelligence: A brief review. In *Analyzing Future Applications of AI, Sensors, and Robotics in Society*; Musiolik, T., Cheok, A., Eds.; IGI Global Scientific Publishing: Hershey, PA, USA, 2021; pp. 1–16. [CrossRef]
67. Ilyas, M. Emerging role of artificial intelligence. *J. Syst. Cybern. Inform.* **2022**, *20*, 58–65. [CrossRef]
68. Cooper, G. Examining science education in ChatGPT: An exploratory study of generative artificial intelligence. *J. Sci. Educ. Technol.* **2023**, *32*, 444–452. [CrossRef]
69. Alt, T.; Ibsch, A.; Meiser, C.; Wilhelm, A.; Zimmer, R.; Berghoff, C.; Droste, C.; Karschau, J.; Laus, F.; Plaga, R.; et al. *Generative AI Models: Opportunities and Risks for Industry and Authorities*; Federal Office for Information Security: Bonn, Germany, 2024. [CrossRef]
70. Kalota, F. A Primer on generative artificial intelligence. *Educ. Sci.* **2024**, *14*, 172. [CrossRef]
71. Banh, L.; Strobel, G. Generative artificial intelligence. *Electron. Mark.* **2023**, *33*, 63. [CrossRef]
72. Kılınç, H.K.; Keçecioglu, Ö.F. Generative artificial intelligence: A historical and future perspective. *Acad. Platf. J. Eng. Smart Syst.* **2024**, *12*, 47–58. [CrossRef]
73. Yildirim, E. A crisis in education: Navigating the technological abyss in architectural design studio. In Proceedings of the 2nd International Cankaya Scientific Studies Congress, Ankara, Türkiye, 28–29 September 2023. Available online: https://www.researchgate.net/publication/375059300_A_Crisis_In_Education_Navigating_The_Technological_Abyss_In_Architectural_Design_Studio (accessed on 30 November 2025).
74. Ceylan, S. Artificial intelligence in architecture: An educational perspective. In *Proceedings of the 13th International Conference on Computer Supported Education, CSEDU, Virtual Event, 23–25 April 2021*; SciTePress: Setúbal, Portugal, 2021; Volume 1, pp. 100–107. [CrossRef]
75. Başarır, L. Modelling AI in architectural education. *Gazi Univ. J. Sci.* **2022**, *35*, 1260–1278. [CrossRef]
76. Çelik, A.T.; Kurtulus, S.; Karabay, E.; Özdemir, S. Integrating AI image generation to first year design studio: “invisible cities” reimagined with AI subtitle. In Proceedings of the ASCAAD 2023, Amman, Jordan, 7–9 November 2023. Available online: https://papers.cumincad.org/data/works/att/ascaad2023_060.pdf (accessed on 15 December 2025).
77. Lukovich, T. Artificial intelligence and architecture towards a new paradigm. *YBL J. Built Environ.* **2023**, *8*, 30–45. [CrossRef]
78. Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C.; Mishkin, P.; Zhang, C.; Agarwal, S.; Slama, K.; Ray, A.; et al. Training language models to follow instructions with human feedback. *arXiv* **2022**, arXiv:2203.02155. [CrossRef]
79. Sukkar, A.W.; Fareed, M.W.; Yahia, M.W.; Mushtaha, E.; de Giosa, S.L. Artificial intelligence Islamic architecture (AIIA): What is Islamic architecture in the age of artificial intelligence? *Buildings* **2024**, *14*, 781. [CrossRef]

80. Hegazy, M.; Saleh, A. Evolution of AI role in architectural design: Between parametric exploration and machine hallucination. *MSA Eng. J.* **2023**, *2*, 262–288. [[CrossRef](#)]
81. Rafsanjani, H.; Nabizadeh, A.H. Towards human-centered artificial intelligence (AI) in architecture, engineering, and construction (AEC) industry. *Comput. Hum. Behav. Rep.* **2023**, *11*, 100319. [[CrossRef](#)]
82. Borglund, C. *Artificial Intelligence in Architecture and Its Impact on Design Creativity: A Study on How Artificial Intelligence Affects Creativity in the Design Process*; KTH Royal Institute of Technology: Stockholm, Sweden, 2022.

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